Sri Lanka Institute of Information Technology



IT3091 Machine Learning

Final Group Submission Report

Weekly Colombo wholesale price prediction and crop-yield forecasting for vegetable planning

Group 2026-AI-16 | Industry Explorer Track

IT24100858 - Wijesinghe D.T.D.

IT24100533 - Danthanarayana D.M.R.

IT24101090 - Jayakody N.D.

IT23361690 - Gunathilake B.M.P.

GitHub repository: https://github.com/DTD-Wijesinghe/IT3091---Machine-Learning-Assignment

Streamlit application: https://uyvp2rwamprdaupnh9sp3r.streamlit.app/

Project video: https://www.youtube.com/watch?v=B44hwI6cwd8

Project summary AgriPlan presents two separate estimates: next-observed-week Colombo wholesale price and crop yield per hectare. The app places both signals in a practical planning interface; it does not promise profit or a guaranteed harvest.

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1. Project overview and assignment alignment

AgriPlan addresses a familiar planning question for Sri Lankan vegetable growers: how to consider expected harvest and near-term market conditions before deciding what and how much to cultivate. The submitted system provides two regression estimates in one interface, while keeping their datasets, targets, and models separate.

The active price pipeline predicts the next observed weekly Colombo-and-suburbs wholesale price in Rs/kg. The separate yield pipeline estimates tonnes per hectare from production records. When a user enters hectares, the app also shows estimated total production. These values are planning signals, not a crop-choice optimizer or a guarantee of income.

The report follows the descriptor's core evidence areas: problem framing, workflow and decision log, data understanding, preprocessing, model comparison, evaluation, recommendation and limitations, reproducibility, and AI-use transparency. The Industry Explorer context is described without treating a review note as a formal institutional endorsement.

2. Problem framing and stakeholder context

Canvas item	**Project definition**
Track	Industry Explorer Track; Sri Lankan vegetable-planning context.
Decision need	Growers and agricultural planners benefit from considering expected yield alongside near-term wholesale-price signals.
Proposed client context	Clean Sri Lanka Programme, as named in the group's project framing; the intended decision users are vegetable farmers and agricultural officers.
Primary decision lens	Production planning: review yield per hectare and the next observed weekly Colombo wholesale price before using local agronomic and market knowledge.
Unit of analysis	Price: crop-market-week record. Yield: crop-district-season-year record.
ML tasks	Two separate supervised regression pipelines, not one multi-output model.
Price target	NextWeekWholesalePrice_RsKg.
Yield target	Yield_t_ha = Production / Extent.
Success evidence	Transparent row counts, appropriate time-aware price evaluation, separate metrics by target, a baseline and alternatives for price, and reproducible saved artifacts.

A review note supplied with the project describes the yield-plus-price concept as potentially useful to farmers and agricultural planners and emphasizes that predictions support, rather than replace, decisions. The project is a student prototype; the app is not presented as an official service of the proposed client context.

3. Data sources and record counts

The active forecasts use local Sri Lankan production and weekly wholesale-price records. The daily Colombo workbook and AGRI-ONE archive are retained in the repository's raw-data area but are not inputs to the current deployed forecast pipelines.

Source	Raw / available data	Prepared or model records	Role in current system
researchData.xlsx	94,755 source rows (2001-2025).	60,529 prepared yield rows; 48,423 train and 12,106 test in the saved 80/20 evaluation.	Active standalone crop-yield pipeline.
Weekly Average Retail Prices of Vegi - 2012 to 2025.xlsx	Weekly W1-W52 workbook; Colombo and suburbs, filtered to wholesale.	15,598 prepared crop-week observations; 15,546 rows with lag and target available; 9,387 train and 6,159 later-period evaluation rows.	Active next-week price pipeline and historical trend.
weatherData.csv and locationData.csv	Daily weather and 27 mapped locations in the group preparation work.	Weather summaries were prepared for earlier group modelling work.	Not features in the current two deployed models.
2018-2025 Daily Wholesale Prices (Colombo).xlsx	Daily Colombo workbook with yearly sheets.	Not merged into the active weekly model target.	Supporting source only; avoids mixing frequencies.
AGRI-ONE.zip	Archive present in Raw Datasets.	Not used in the active app models.	Not included in current price or yield predictions.

The app currently shows 15,598 weekly price records, 26 available price crops, 60,529 prepared yield rows, and a latest stored weekly price date of 3 March 2025. The number of model-ready price rows is lower because records without a usable previous-price feature or next-observed-week target are excluded.

4. Data dictionary

Pipeline	Field	Meaning / use
Weekly price	Crop; Market	Crop name and Colombo-and-suburbs wholesale market label.
Weekly price	WeekDate; Year; WeekNumber; Month	Calendar fields for the observed price week.
Weekly price	WholesalePrice_RsKg	Current observed weekly price; model input.
Weekly price	PreviousWeekWholesalePrice_RsKg	Previous observed weekly price; model input.
Weekly price	NextWeekWholesalePrice_RsKg	Next observed price record; prediction target.
Yield	District; Season; CropCategory; Crop; Year	Location, season, crop grouping, crop, and year features.
Yield	Extent	Cultivated hectares supplied as a yield-model feature.
Yield	Production	Source production amount used to calculate historical yield.
Yield	Yield_t_ha	Production divided by Extent; model target in tonnes per hectare.

5. Workflow and member contributions

The current deployment is built from two parallel pipelines. The earlier production and weather notebooks remain part of the group's data-preparation evidence, while the app reads the committed weekly-price and separate crop-yield artifacts.

Pipeline	Raw source		Prepared data		Model		App result
Price	Weekly workbook	->	Crop-week rows and current/previous price	->	Linear Regression	->	Next observed weekly wholesale price (Rs/kg)
Yield	researchData.xlsx	->	60,529 rows; district, season, crop, year, extent	->	Random Forest	->	Yield (t/ha) and estimated production

The price model uses crop, market, year, week number, month, current price, and previous observed-week price. The yield model uses district, season, crop category, crop, year, and extent. The app applies the separately entered hectares to estimate total production.

Member	Main contribution reflected in the repository
IT24100858 - Wijesinghe D.T.D.	Production-data preparation and a separate crop-yield preprocessing/model workflow; yield target construction and hectares-based estimate.
IT24100533 - Danthanarayana D.M.R.	Weather/location validation and time-window preparation in the group data workflow.
IT24101090 - Jayakody N.D.	Weekly wholesale-price EDA, crop mapping, and price preprocessing.
IT23361690 - Gunathilake B.M.P.	Weekly price-model comparison, evaluation artifacts, and model handoff to the app.

6. EDA and preprocessing decisions

Area	Observed / applied	Reason
Production source	94,755 source rows; numeric year, extent and production fields checked.	Preserves the source meaning and supports a reproducible target calculation.
Yield preparation	Remove unusable fields, require positive extent and non-negative production, calculate Production / Extent, and retain values at or below the 99.5th percentile.	Creates valid tonnes-per-hectare targets and limits the effect of extreme ratios.
Weekly price source	Reshape W1-W52 into crop-week observations and retain wholesale observations.	Keeps the target at the source's weekly resolution.
Price lag / target	Use the previous observed price record as a feature and the next observed price record as target.	Missing source weeks mean the next recorded interval can exceed seven days; it is not filled with invented prices.
Missing and invalid values	Drop rows that cannot supply the required model features or target.	Avoids fabricating price or production observations.
Categorical and numeric fields	One-hot encode categories; scale numeric columns for the price Linear Regression pipeline; use a tree-based yield regressor.	Matches preprocessing to the estimator and input types.
Weather and location work	Validate location IDs and use only prior weather windows in the group preparation notebook.	Shows time-aware feature design; these variables are not used by the current deployed models.

7. Model strategy and evaluation

The price and yield targets have different units and are evaluated independently. The price pipeline compared a dummy baseline, Linear Regression, Random Forest, and Gradient Boosting. The saved yield pipeline is a separate Random Forest regressor evaluated on a random 80/20 row split.

Weekly price model	Rows	MAE (Rs/kg)	RMSE (Rs/kg)	R^2
Linear Regression	15,546	27.24	48.33	0.906
Random Forest	15,546	31.67	72.44	0.789
Gradient Boosting	15,546	32.48	80.09	0.742
Dummy baseline	15,546	118.87	190.13	-0.456

For the price evaluation, records through 2019 form the training period and 2020 onward is the reported later-period comparison (6,159 rows). The best reported model is Linear Regression. The scores describe this historical evaluation split, not a promise of future accuracy.

Yield model	Records	Train / test	MAE	RMSE	R^2
Random Forest Regressor	60,529	48,423 / 12,106	1.99 t/ha	3.10 t/ha	0.676

The yield R^2 of 0.676 is from a random row split, so it summarizes fit on held-out rows from the same prepared dataset and should not be described as a future-season accuracy percentage. R^2 is a regression score, not classification accuracy.

8. AgriPlan application and recommendation

The public Streamlit prototype has Overview, Weekly price forecast, Crop-yield forecast, and Data coverage pages. The Overview page lets a user select a crop and review its stored weekly trend. The price page asks for a crop, observation week, current price, and previous-week price from the same Colombo wholesale market. It then displays a next-week estimate and expected movement.

The yield page accepts crop, season, district, production year, and cultivated hectares. It displays estimated yield in t/ha and estimated production calculated from yield and hectares. The Data coverage page lets the user filter recent price observations by crop and shows the coverage counts and latest stored date.

Practical use Use the price estimate as one market signal alongside current market checks. The stored series ends on 3 March 2025, so a current forecast requires the user to enter up-to-date current and previous-week market prices.

9. Responsible use and scope

AgriPlan is a decision-support prototype for discussion and planning. It does not calculate profit, recommend a guaranteed best crop, or replace agricultural officers, farmer experience, or current market information. The price model represents the supplied Colombo-and-suburbs wholesale series; it should not be labelled a national price forecast.

The price estimate can be affected by demand, supply, imports, transport costs, inflation, festivals, weather events, and market disruptions that are not model inputs. The yield estimate can be affected by soil, irrigation, seed variety, fertilizer, pests, and farm practices not captured in the training table. Yield metrics use a random split, so a future-year validation would provide stronger evidence about year-to-year generalization.

The training files contain aggregate crop, season, district, weather, and market variables rather than personal farmer identities. The system should continue to present units, data coverage, and the decision-support caution wherever predictions are displayed.

10. Reproducibility and submission evidence

The GitHub repository contains the raw data folder, group notebooks 01-04, the standalone crop-yield notebook 07, saved preprocessing outputs, model artifacts, metrics, predictions, and the app entry point. Notebooks 01 and 02 document the group production/weather work; 03 and 04 cover the weekly price work; 07 independently creates the current 60,529-row yield dataset and its model outputs.

For a clean price-workflow rerun, confirm that the source workbook's Commodity field is mapped to the Crop field expected by the prepared table, and that the model notebook consumes the saved weekly output. The current committed app artifacts and displayed coverage are populated; the run-order check is included so a future rerun reproduces the same weekly series.

Notebook	Purpose	Main saved output
01_IT24100858_Wijesinghe_DTD_Production.ipynb	Production-data preparation.	Preprocessed Datasets/01_production_preprocessed.csv
02_IT24100533_Danthanarayana_DMR_Weather.ipynb	Weather and location preparation.	Preprocessed Datasets/02_weather_preprocessed.csv
03_IT24101090_JAYAKODY_ND_Price_Weekly_Only.ipynb	Weekly wholesale-price preparation.	Preprocessed Datasets/03_price_preprocessed.csv
04_IT23361690_Gunathilake_BMP_Weekly_Only.ipynb	Weekly price-model comparison and evaluation.	weekly_price_model.joblib and weekly_price_* metrics/predictions
07_Wijesinghe_DTD_Crop_Yield_Preprocessing.ipynb	Separate crop-yield preprocessing and Random Forest model.	Wijesinghe.D.T.D-Crop Yield Preprocessing/ and Wijesinghe.D.T.D-Crop Yield Outputs/

Descriptor evidence	Where this report / repository presents it
Problem framing canvas	Section 2.
Workflow diagram and decision log	Section 5 and Appendix A.
Data dictionary, EDA, and data quality	Sections 3, 4, and Appendix B.
Preprocessing and feature log	Section 6 and Appendix C.
Model comparison and evaluation	Section 7; saved price comparison, price metrics, and yield metrics.
Recommendation and limitations	Sections 8 and 9.
Reproducibility and AI transparency	Section 10 and the repository README; AI-use statement in section 11.
Final group video and individual reports	provided
Industry Explorer review evidence	provided

Appendix A. Decision log

Decision	Choice	Reason
Forecast resolution	Weekly price; seasonal yield.	Matches the two separate source and model pipelines.
Price geography	Colombo and suburbs.	Reflects the weekly workbook; does not imply national wholesale coverage.
Main price source	Weekly workbook, wholesale rows.	Preserves source frequency and market type.
Price horizon	Next observed price record.	Uses the next available source observation without filling missing weeks.
Yield target	Production / Extent in t/ha.	Makes the target comparable across cultivated area sizes.
Yield inputs	District, season, crop category, crop, year, extent.	Matches fields used by the saved yield pipeline and app inputs.
Price evaluation	Train through 2019; report 2020 onward.	Preserves chronological order for the next-week target.
Yield evaluation	Random 80/20 split.	Matches the saved notebook evaluation; interpreted as row holdout evidence.
Separate models	Linear Regression for price; Random Forest for yield.	Avoids mixing different targets, frequencies, units, and data sources.

Appendix B. EDA insight log

Observation	Interpretation for the project
15,598 weekly wholesale crop-week observations; 26 crops in the app.	The source retains weekly detail; rows without lag/target are not used for model fitting.
Latest stored price date: 3 March 2025.	Current price forecasts need current market inputs from the user.
60,529 valid yield rows from 94,755 raw records.	Cleaning retains a large, multi-year district/crop/season table for the standalone yield model.
Price errors are measured in Rs/kg; yield errors in t/ha.	Report metrics separately and do not combine raw errors across units.
Market prices vary over time and across crops.	Use chronological price evaluation and interpret the score within its market and date coverage.

Appendix C. Preprocessing and feature log

Topic	Handling
Yield numeric validation	Convert year, extent, and production to numeric; remove missing or invalid values.
Yield construction	Require Extent > 0 and Production >= 0; calculate Production / Extent.
Extreme yield ratios	Retain rows up to the 99.5th percentile in the standalone yield preparation.
Price reshape	Convert weekly W1-W52 columns to one observation per crop and week.
Price calendar fields	Derive week date, year, ISO week number, month, and market label.
Price lag and lead	Within crop, shift observed prices to construct previous-price feature and next-observed target.
Missing model rows	Drop rows lacking the required features or target; do not impute market prices.
Categorical model inputs	One-hot encoding with unknown-category handling.
App outputs	Yield per hectare and estimated production = predicted yield × user-entered hectares.

Appendix D. Raw dataset fingerprints

SHA-256 values below were calculated from the raw files in the repository's Raw Datasets folder. They allow a later rerun to check whether the source files are byte-for-byte the same.

Raw file	SHA-256
researchData.xlsx	4b2f922b03d93c0d7a46b68aba1546db9278767a4ab910b950053373229c2cb1
Weekly Average Retail Prices of Vegi - 2012 to 2025.xlsx	ab5303299e29ccbbcd495cea646fbe00f06b88adcd0c5a6d4a31d12bea5d16fa
weatherData.csv	cc75e8c1cdc82d5ee0f2e75e80490fecc050247619455bfb751bcff80b90a1ee
locationData.csv	912c02712a422632650f510ce00bb4a5754aa823384fd593a021616eb5d01a7e
2018-2025 Daily Wholesale Prices (Colombo).xlsx	a48cc494eeb10258331de414ae0e4323e7cc59d5066f435b8d56c4c12a79311f
AGRI-ONE.zip	bdfeb77a52f7e8311d45a7204f84e10f26d8b1108494e311bc19a4c775e4e2ba

11. AI-use statement

Generative AI tools were used to support explanation and debugging.

Prompts

- Explain how to prepare weekly wholesale price data for a next-week prediction task, and identify checks needed to avoid using future information.
- Help debug this Python error in my preprocessing or model evaluation code. Explain the cause and suggest a correction.